

Browsing Large Graphs with Tile Pyramids and Sleeve Routing in the Browser

Category: System

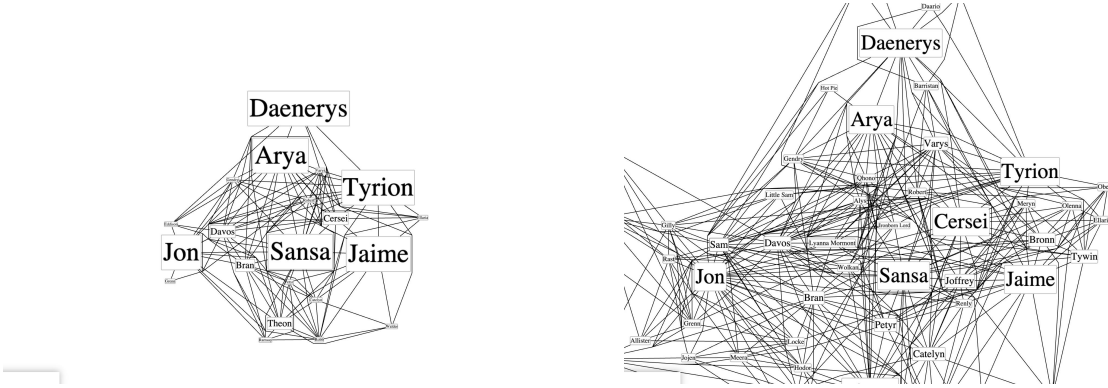


Figure 1: Two adjacent pyramid levels of the Game of Thrones graph rendered by the WebGL renderer of GJS with the sleeve router. The coarser level is on the left, the finer one on the right; both panels are cropped to the same world window. For each level a new graph is assembled from a PageRank-ranked prefix of the nodes, and edges are routed against the obstacles introduced at that level.

ABSTRACT

We present a new way to visualize a large graph in the style of online geographic maps. The method builds a *tile pyramid* for *semantic zoom*: at every zoom level the labels of the highest-ranked nodes remain readable, just as the names of major geographical features stay readable on those maps.

The edges are routed by a method we call *sleeve routing*, which searches the dual graph of a Constrained Delaunay Triangulation to select a sequence of triangles through the free space, then applies the funnel algorithm to compute a shortest path inside the selected sleeve. We apply several heuristics to speed up the routing.

We implemented our approach in the WebGL renderer of GJS (a placeholder name used during double-blind review), an open-source TypeScript library for graph visualization in web browsers, with the entire pipeline running client-side, without a dedicated server. Our benchmark suite contains nine graphs with up to 32,768 nodes and 236,978 edges, and measures browser-side parsing, layout, routing, and tile-pyramid construction. The renderer’s demo can be seen at <https://gjs-sleeve.surge.sh/>; the sources and packages are described under Supplemental Materials.

Index Terms: Graph visualization, edge routing, Constrained Delaunay Triangulation, tiling, WebGL.

1 INTRODUCTION

Visualizing graphs in web browsers without a dedicated server presents practical challenges: long-running computations can block the main thread unless offloaded to workers, browser memory is limited, and users expect the smooth pan-and-zoom experience familiar from online maps. GJS is an open-source TypeScript library that addresses these challenges by combining efficient layout algorithms, a novel edge routing method, and a tiling scheme with semantic zoom.

GJS is consumed as a set of NPM packages and renders graphs using WebGL through the deck.gl framework [2]. It supports the layered Sugiyama scheme [46], Pivot MDS [15], and IPSep-CoLa [22], with edges routed around node obstacles. Throughout this paper our typical example is a social network laid out by IPSep-CoLa; the methods we describe, however, are agnostic to the layout algorithm

and apply to any node placement in which the nodes do not overlap. The graph loading pipeline builds a hierarchy of tiles—analogueous to map tiles—so that at any zoom level only the entities in visible tiles are drawn. The capacity parameter guides pyramid construction but is not a hard bound on rendered tile occupancy. Tiles are supplied locally to the browser through the standard `TileLayer` of the `@deck.gl/geo-layers` package, giving smooth pan and zoom. A live demo of the WebGL renderer is available online¹.

This paper makes two main contributions:

1. A *tiling scheme* for large graph visualization that builds a hierarchical tile pyramid, reduces the number of visible entities using PageRank-based filtering and edge-clip bundling, and integrates with the WebGL renderer for interactive pan and zoom.
2. A *sleeve routing* algorithm that selects triangle sequences on the dual graph of a Constrained Delaunay Triangulation using per-source batched Dijkstra searches, then applies the classical funnel algorithm to find shortest paths within the selected sleeves. Sleeve selection is heuristic, not globally shortest-path optimal.

2 RELATED WORK

Graph visualization tools. Graphviz [26, 5] is a long-standing graph drawing tool supporting multiple layout algorithms, including the Sugiyama method and Scalable Force-Directed Placement [7]; it does not support tiling or WebGL rendering. MSAGL [41] is a desktop layout engine that did not have the features described here. On the web, Sigma.js [8] and Cytoscape.js [24] provide interactive graph rendering, but rely on straight-line or generic spline edges without obstacle-avoiding routing and without a pyramid of precomputed tiles. ReGraph [6] uses WebGL to render large graphs with straight-line edges and supports interactive cluster expansion but not tiling. Cosmograph [1] uses GPU-accelerated force-directed layout for million-node graphs with straight-line edges, without tiling. GraphMaps [40] introduced the tile-pyramid and node-ranking scheme, but it runs only on Windows, ships raster tiles,

¹<https://gjs-sleeve.surge.sh/>

and reuses one global edge routing that is merely clipped per tile; GJS brings the idea to the browser with vector tiles and per-level sleeve routing. Cornac [43] handles huge graphs with tiles and edge bundling [35, 33] but requires a multi-server backend. Earlier navigation schemes for large graphs include ASK-GraphView [12], which organizes nodes into a hierarchical cluster tree, and topological fish-eye views [25], which distort a multilevel layout [28, 34] under the focus; neither targets browser-side tile pyramids or per-level edge routing. Hierarchical edge bundling [32] and force-directed bundling [33] offer alternative clutter-reduction strategies; but neither method guarantees that bundled edges avoid node obstacles.

Shortest paths and edge routing. Our sleeve algorithm walks the dual of a Constrained Delaunay Triangulation [45] and relies on the classical funnel algorithm for extracting a geodesic from a triangle strip [38, 16, 27, 4]. The optimal algorithm for Euclidean shortest paths among polygonal obstacles [31] and the recent $O(m \log^{2/3} n)$ single-source shortest-path algorithm that breaks the sorting barrier [21] are remarkable theoretical results. We do not implement or benchmark these algorithms; our engineering choice is to reuse a CDT and conventional graph searches. Graphviz builds the full visibility graph for edge routing in force-directed layouts, which has $O(n^2)$ edges. For Sugiyama layouts, it uses the funnel algorithm [19] but requires a simple containing polygon. yWorks [11] offers “Organic edge routing” based on force-directed simulation. The approach of Dwyer and Nachmanson [23] routes on a Yao-graph spanner with local shortcutting. For orthogonal layouts, the libavoid framework of Wybrow, Marriott, and Stuckey performs obstacle-avoiding connector routing with incremental updates [47].

CDT-based pathfinding. The closest precedent to our routing pipeline is the CDT-and-funnel pathfinder of Demyen and Buro [18], originally designed for game-AI units of nonzero radius. They run admissible A* search on the CDT dual (with a reduced-graph variant that collapses corridor chains into single abstract edges) to recover provably shortest paths. Their search is, however, per-query: the bounds depend on the start and goal, so it cannot share work across the edges with the same source as our per-source Dijkstra does. For the graphs we target, sharing work across queries is attractive. We trade a global optimality guarantee for batched search; we have not benchmarked their pathfinder directly, so we make no quantitative speed comparison with it.

3 TILING

The input to tiling is a laid-out graph together with routed edges. We use sleeve routing, Section 4, both for the input graph and for the coarser levels; routes at different levels are computed independently, so continuity across level changes is not guaranteed. The output is a sequence of $Z+1$ levels: level Z is the finest, and level 0 is the coarsest. Each level is a collection of tiles indexed by integer (x, y) grid coordinates.

A tile is an axis-aligned square together with its *tile data*: nodes, edge labels, edge arrowheads, and *edge clips*. An edge clip is a curve confined to the tile square, paired with the non-empty list of graph edges whose clipped routes have the same quantized endpoint pair including endpoints inside the tile. Bundles are intended to capture stretches on which several routes travel close together inside a common sleeve; we represent each such stretch as a single clip with an attached edge list, rather than as one clip per edge. All tiles at level z have side length $\ell_z = \ell_0 / 2^z$, where ℓ_0 is the root tile’s side length. Each subdivision produces four congruent square sub-tiles of half the parent’s side length.

Pseudocode 1 describes the algorithm building the tile pyramid.

How the index of the finest layer, Z , is chosen. We treat the tiles differently when calculating Z and when preparing for rendering. There is no element filtering in the former case: all graph entities are represented in each level. We continue increasing Z ,

Algorithm 1 BUILD_TILE_PYRAMID($G, \mathcal{C}, \mathcal{M}_{\max}$)

```

1:  $T_0 \leftarrow$  single square tile of side  $\ell_0 = 2^{\lceil \log_2 \max(w, h) \rceil}$  centered on
    $G$ ’s bounding box of width  $w$  and height  $h$ , holding all of  $G$ 
2:  $(\bar{w}, \bar{h}) \leftarrow$  mean original node width and height
3:  $levels \leftarrow [T_0]$ ;  $z \leftarrow 0$ ;  $used \leftarrow |T_0| \triangleright |T| = \text{element count of tile } T$ 
4: while some tile in  $levels[z]$  exceeds  $\mathcal{C}$  elements do
5:    $next \leftarrow$  empty  $2^{z+1} \times 2^{z+1}$  tile grid
6:   for all tiles  $T \in levels[z]$  do  $\triangleright$  per-tile inner loop
7:     split  $T$  into four congruent square sub-tiles of  $next$   $\triangleright$  Section 3.2
8:    $used +=$  number of new elements added to  $next$ 
9:   if  $200 \cdot used > \mathcal{M}_{\max}$  then  $\triangleright$  memory budget; checked mid-split
10:    discard  $next$  and break pyramid growth
11:   end if
12: end for
13:  $levels[z+1] \leftarrow next$ ;  $z \leftarrow z+1$ 
14: if  $\ell_z \leq 10 \min(\bar{w}, \bar{h})$  then
15:   break  $\triangleright$  retain this completed level
16: end if
17: end while
18:  $Z \leftarrow z$ ;  $V_Z \leftarrow V(G)$ ;  $s_v^{(Z)} \leftarrow 1$  for all  $v \in V_Z$ 
19:  $V_{PR} \leftarrow$  nodes of  $G$  sorted by descending PageRank
20: for  $z = Z-1$  down to 0 do
21:    $(V_z, \{s_v^{(z)}\}) \leftarrow \text{SELECTTOPKWITHADAPTIVESCALE}(V_{PR}, \lceil |V|/2^{Z-z} \rceil)$   $\triangleright$  Section 3.1
22:    $E_z \leftarrow \{uv \in E(G) : u, v \in V_z\}$ 
23:   build a CDT from the inflated polylines of  $V_z$  at scales  $\{s_v^{(z)}\}$ 
24:   route every edge  $e \in E_z$  on this CDT
25:   rebuild the tiles of  $levels[z]$   $\triangleright$  Section 3.2
26: end for
27: return  $levels$ 
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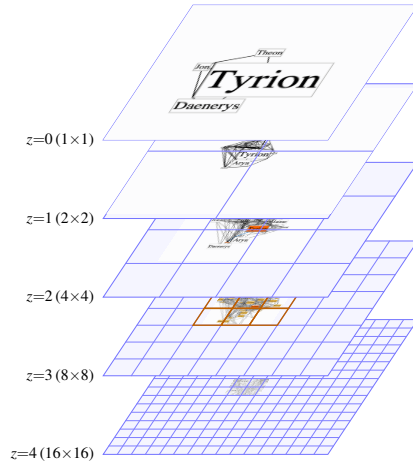


Figure 2: Tile grids and viewports at five pyramid levels. Each screenshot uses its level’s display sizes and zoom; the blue grid shows the full world extent. The orange outlines mark the six tiles intersected by the viewport at level 3.

generating new levels, and splitting tiles, until a stopping condition is met. Then, while keeping the finest level intact, we rebuild the tiles of levels $0, \dots, Z-1$ for rendering.

Let w_v and h_v be the width and height of node v ’s axis-aligned bounding box in the original layout, before padding or level-

dependent scaling. For the non-cluster nodes V , define

$$\bar{w} = \frac{1}{|V|} \sum_{v \in V} w_v, \quad \bar{h} = \frac{1}{|V|} \sum_{v \in V} h_v.$$

These means are computed once over the full graph, not separately for each tile or PageRank prefix. The tile side length ℓ_z is measured in the same world-coordinate units.

We start from $Z = 0$; if the root tile already meets the capacity condition, no subdivision is needed. Otherwise we grow the pyramid one level at a time and stop as soon as any of three conditions is met:

1. every tile at the new level contains at most $\mathcal{C}$ elements, with default $\mathcal{C} = 500$;
2. the completed new level z satisfies $\ell_z \leq 10 \min(\bar{w}, \bar{h})$. Equivalently, the side length is at most ten times both the mean node width and the mean node height; equality triggers the stop. This level is retained as the finest level, $Z = z$;
3. while generating the tiles of the new finest level, the running total of stored tile elements summed over all levels, multiplied by 200 bytes per element, exceeds the memory budget $\mathcal{M}_{\max}$, with default 4 GiB (2^{32} bytes); the partial level is then discarded and Z is set to the previous finest level.

The third condition is checked incrementally inside the level-build loop. This is a storage estimate, not a bound on the browser’s actual heap: it does not account for all object overhead, edge-membership lists, routing workspace, or GPU buffers, and is not rechecked during the subsequent per-level rebuilding. On every benchmark graph in Table 1 the first condition fires; the second and third are defensive guards that bound depth on hypothetical inputs whose finest level cannot be packed within $\mathcal{C} = 500$.

Three phases. Once Z and the tile squares are fixed, each coarser level $z = Z-1, Z-2, \dots, 0$ is built in three phases:

1. pick the nodes of level z with PageRank and possibly enlarge them, Section 3.1;
2. reroute the edges of G_z around the resulting obstacles, Section 4;
3. recompute the per-tile edge clips on level z , Section 3.2.

Figure 1 shows two adjacent pyramid levels built by this procedure; Algorithm 1 summarizes the whole pipeline.

3.1 PageRank-Guided Level Construction

Before filling the level’s tiles for rendering we run PageRank [42] on G once, and sort all nodes of the graph, V , by descending rank. Write $k = Z - z$ for the level’s depth below the finest level. The candidates for a level of depth k are the nodes of the *prefix* of the sorted array of length $\lceil |V|/2^k \rceil$. Let V_z be the nodes accepted from this prefix by the overlap filter below. The level graph is $G_z = (V_z, E_z)$, where $E_z = \{uv \in E(G) : u, v \in V_z\}$; an edge is excluded if either endpoint is dropped.

Node positions are inherited from the original layout and never move: across levels we change only each node’s display size.

We add nodes to a level of depth k by walking the level’s prefix in rank-descending order. The top-ranked node is scaled by 2^k . We keep the node scales non-increasing, but as large as possible, while making sure that each next node does not overlap the previously selected nodes, with the routing padding included in the overlap test. Each accepted scale lies between 1 and the preceding accepted scale. If even at scale 1 the candidate’s padded bounding box would overlap an already-accepted node’s padded scaled box, the candidate is dropped.

Dropping a relatively high-ranked candidate at a coarse level is a deliberate compromise: we trade the inclusion of that candidate at this level for the invariant that the rendered nodes at every level remain pairwise disjoint. The cost is small in practice. Consider Arya’s node in the *Game of Thrones* graph: as Figure 2 shows, it is dropped from the top layer because a higher-ranked node, Tyrion, has covered the available region. The user still sees that part of the graph as occupied, and Arya’s individual node appears at a finer level where its box fits. The candidate prefixes are nested, but the accepted sets V_z need not be: overlap filtering is performed independently at each level. All nodes are present at the finest level.

Spatial-hash overlap test. A naive overlap test against every previously accepted node would dominate the cost of the filter. We instead maintain the accepted boxes in a uniform spatial hash whose cell size is chosen so that any accepted padded box, and any candidate’s padded box at the level’s maximum scale, have width and height at most the cell size. Because of this choice, two such boxes can overlap only if their center cells differ by at most one step on each axis: testing a candidate therefore reduces to scanning the accepted boxes registered in the candidate’s own cell and in the eight cells immediately around it.

The hash brings little benefit at the coarsest levels, where the prefix is short and even a naive scan over the few accepted boxes is cheap; it pays off at finer levels, where the prefix grows geometrically. The number of cells examined per candidate is constant, but their occupancy is not bounded; a query costs time proportional to the number of boxes in those cells, and the filter can still take quadratic time in the prefix length in the worst case. The spatial hash is therefore an acceleration heuristic rather than a constant-time overlap guarantee.

3.2 Splitting Edges between Tiles

We describe edge routing later in Section 4, but for the purpose of this paragraph it is enough to know that each curve is a polyline.

The edge-clip generation works as follows. Consider a tile and an edge clip contained in the tile. Let the *midlines* of a tile be the horizontal and vertical segments through its center; they split the tile into four congruent square sub-tiles. Cutting the curve with the midlines yields a set of sub-clips, each contained in one sub-tile. For isolated boundary contacts, splitting at tangencies as well as crossings ensures that each sub-clip meets the boundary only at its endpoints. Segments lying along a midline can be assigned to either incident sub-tile using a consistent tie-breaking rule. Every split reuses the parent clip’s boundary crossings, and only the curve-vs-midline intersections need to be computed; no full curve-vs-square clipping is required. Figure 3 illustrates one such split.

When we grow the pyramid in line 7 of Algorithm 1, each edge clip is already contained in its parent tile, so subdivision starts from that parent’s square. When we prepare the tiles for rendering in line 25 of Algorithm 1, we have the rerouted graph G_z , so for each edge of G_z we start the subdivision from the root tile’s square.

Bundling edge clips. To reduce the number of elements rendered in a tile, all clips within a tile that share the same unordered pair of quantized endpoints are bundled into a single clip whose attached edge list accumulates every contributing edge. Each endpoint coordinate is rounded to the nearest multiple of 10^{-6} world units. The bundled geometry is taken from the first such clip encountered during splitting, and the choice is consistent within each level. This endpoint-based test does not bound the deviation between the interiors of the original curves and their representative. Table 1 reports the resulting maximum number of elements rendered in a single tile across the entire pyramid for a range of graphs.

4 SLEEVE ROUTING ON THE CDT

To prepare for routing given the graph layout, each node is covered by a padded obstacle polygon. We compute a Constrained Delaunay

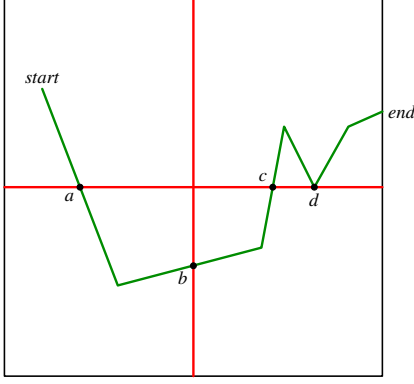


Figure 3: An edge clip, drawn as a polyline, from *start* to *end* inside a tile. The two red midlines split the tile into four congruent square sub-tiles and meet the polyline at the four points *a*, *b*, *c*, *d*; at *d* the polyline touches the horizontal midline without crossing it. Cutting the polyline at these points yields five sub-clips $[start, a]$, $[a, b]$, $[b, c]$, $[c, d]$, $[d, end]$, each contained in one sub-tile and meeting that sub-tile's boundary only at its endpoints.

Table 1: Maximum number of elements rendered in a single tile across the whole pyramid for several graphs (capacity $\mathcal{C} = 500$; Z chosen by the natural stops of Section 3). The capacity $\mathcal{C}$ only guides the choice of Z ; we have no tight analytical bound on the actual rendered per-tile element count, which is therefore reported empirically. “At level” is the level on which the maximum tile was found. In the worst cases, *ca-HepPh* and *ca-CondMat*, the densest tile holds approximately $5.9\text{--}7.3\times$ the nominal capacity $\mathcal{C}$.

Graph	$ V $	$ E $	Levels	Max/tile	At level
gameofthrones	407	2 639	5	324	4
composers	3 405	13 832	8	749	3
ca-GrQc	5 242	28 968	8	359	4
ca-HepTh	9 877	51 946	8	1 836	3
facebook_combined	4 039	88 234	9	928	5
ca-HepPh	12 008	236 978	9	3 643	3
ca-CondMat	23 133	186 878	8	2 949	3
deezer_europe	28 281	92 752	8	2 402	3
delanay_n15	32 768	98 274	8	283	7

Triangulation $\mathcal{T}$ on the set of these polygons [17, 20]. The *dual graph* D of $\mathcal{T}$ has one node per triangle and one edge for each pair of triangles sharing a side. The weight of an edge is the Euclidean distance between the centroids of its triangles.

4.1 Routing inside a Sleeve

We split routing an edge into two subproblems: a *combinatorial* one—which sequence of triangles to cross—and a *geometric* one—which exact polyline through that sequence is shortest. The classical funnel algorithm computes the geometric step by pulling the path taut inside the polygon formed by the sleeve triangles [16, 30].

To obtain the sleeve we run a shortest-path search on D , where triangles inside obstacles other than the source and target obstacles are excluded. The natural per-edge instantiation is A^* [29] with the straight-line distance to t as heuristic. When a node has many incident edges, however, running independent searches is wasteful. We instead group edges by source s and run a batched Dijkstra search per source. The source obstacle is traversable; target triangles are terminal destinations, not transit triangles for routes to other targets. Sleeves are recovered by following parent pointers. This gives a speedup over A^* on the social graphs tested; for example on Game of Thrones, with 407 nodes and 2 639 edges, it cuts routing time by

$\sim 23\%$. Table 3 reports the gain across the full benchmark suite.

We can improve even more. Since the query edges are undirected on D , each can be served from either endpoint, so the number of distinct roots is in turn reducible by reorienting query edges to minimize that count. Let the *demand graph* H be the graph whose vertices are the graph nodes and whose edges are the query edges still to be routed. A set of roots R is feasible iff every demand edge has at least one endpoint in R , that is, R is a vertex cover of H , and the smallest such R is a minimum vertex cover of H . The problem of finding a minimum vertex cover is NP-hard [36], so we approximate it with the standard greedy maximum-degree rule: while H still has an edge, pick a vertex of maximum degree in H , add it to R , and delete it together with its incident edges. Implemented with degree buckets, the procedure runs in linear time. Each demand edge is then routed from the first of its endpoints selected by the greedy procedure; the original direction is restored for rendering. Self-loops are handled at their own node. To our knowledge no prior edge router exploits a vertex cover of the demand graph in this way; many-to-many shortest-path frameworks [37] also amortize per-source searches, but they take the source set as given and do not reduce it through a covering argument. Applying this heuristic reduces the number of source groups on Game of Thrones from 331, one per distinct source, to 198, and on the composers graph with 3 405 nodes and 13 832 edges from 2 645 to 1 281, shortening the routing pass by $\sim 28\%$ and $\sim 31\%$ respectively.

In addition, before launching the sleeve search for an edge we try routing the straight line segment between the source and the target by walking the CDT. If it avoids all non-endpoint obstacles, no graph search is needed for that edge.

Route quality. To assess route quality, we compared the sleeve router with the optimum on the full visibility graph of the padded obstacles, on the Game of Thrones layout. The visibility-graph optimum is the shortest taut polyline around the obstacles; we computed it by all-pairs Dijkstra over the polygon corners and the node centers. This is a lower bound for routes with the same endpoints and obstacle constraints. We compare raw center-to-center polylines, before node-boundary and arrowhead trimming. On the 407-node, 2,639-edge layout the sleeve router has total arc length 714,316 world-coordinate units against 695,107 for the visibility-graph optimum, a ratio of $1.028\times$. The worst per-edge ratio is 1.37. The CDT stretch bound $4\pi\sqrt{3}/9 \approx 2.42$ [14] concerns paths on triangulation edges, not centroid-weighted dual searches, and is not an approximation guarantee for our router. These measurements cover one layout only.

Edge hint. We explored a routing variant in which the path generated for an edge on one level provides a hint for routing the same edge on an adjacent level, increasing the visual stability of the routes between levels. Its shortcoming is that the search cannot be batched per source as the Dijkstra-tree variant above is, which impacts performance.

4.2 Vertex Collapse at Source and Target

We improve the sleeve’s geometry in the situation depicted in Figure 4. The remedy is to *collapse* corner-hugging chain vertices to the corresponding endpoint. Walking the right chain of the sleeve outward from s , we find the first vertex of the source obstacle at which the chain makes a right turn, and replace every source-owned vertex on that chain up to and including this one by s . The left chain is processed mirror-symmetrically, with left turns instead of right, and the target end likewise. Diagonals whose two endpoints coincide after collapse are dropped, and the left/right orientation of each surviving diagonal is preserved from the uncollapsed geometry. The collapsed vertex in Figure 4(a) is marked in yellow.

After collapse, the funnel sees a wide opening at each endpoint, as in Figure 4(b), and selects a shorter exit direction. A final arrowhead-trimming step clips the path at the node boundary curves. Collapse

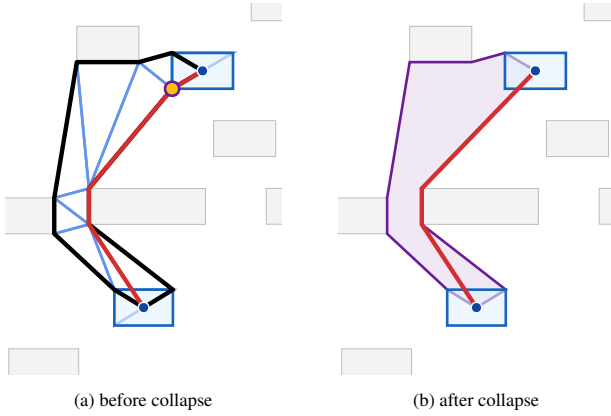


Figure 4: Effect of vertex collapse on the LORAS-RENTLY edge of the Game of Thrones graph. Endpoints are shown as blue dots; their padded obstacles are highlighted in light blue. (a) The CDT triangles forming the sleeve are drawn in cornflower blue and the funnel path in red; the thick black polyline is the perimeter of the non-collapsed sleeve. The yellow dot marks the source-obstacle vertex that gets collapsed. (b) The sleeve after collapse, drawn as a single purple polygon with the source/target obstacles merged into the corresponding endpoints, and the funnel path (red) through it. Collapsing widens the funnel opening at the endpoint and removes the spurious detour around the obstacle corner.

might cause the resulting path to overlap nodes other than the source and the target. Thus the default post-collapse routes do not carry a strict obstacle-avoidance guarantee; we have not quantified the frequency or extent of these overlaps.

The implementation also offers an optional pass that fits a Bezier segment into each polyline corner to render edges as smooth curves. It is disabled by default for performance: the polyline rendering is fast and visually acceptable, so we trade the smoother curves for routing speed.

5 RENDERING WITH DECK.GL

The output of the build pipeline of Section 3 is consumed in the browser by the `TileLayer` of `@deck.gl/geo-layers` [10]. `TileLayer` is the standard component used by online tile-map renderers: it tracks the viewport, determines which tiles are needed at the current zoom, and asynchronously requests them through a user-supplied `getTileData` callback, caching the results across frames so that subsequent pans within the cache avoid tile-data retrieval, although visibility selection and rendering still run.

GJS integrates with this component at construction time by exposing the precomputed pyramid as the `TileLayer`’s zoom range, with the coarsest pyramid level mapped to the layer’s lowest zoom and the finest level to the highest. As the user pans and zooms, `TileLayer` resolves the viewport to an integer zoom and a set of axis-aligned tile indices and, for each tile it needs to display, calls `getTileData`. The renderer translates each such request back into a tile of the corresponding pyramid level and returns its precomputed nodes, edge clips, edge labels, and arrowheads. The mapping is one-to-one, so no rerouting or tile subdivision is performed at view time: panning shifts the visible window over the tile cache, and zooming selects tiles from the corresponding level. The renderer uses `TileLayer`’s no-overlap refinement strategy, not an animated cross-fade between levels.

The rendered geometry depends only on the zoom level and does not change with the viewport, except for the clipping to the visible window. Within each level, the displayed nodes are the ones chosen for that level by the construction of Section 3.1, and the displayed edges use the sleeve routes and bundled edge clips computed

Table 2: End-to-end loading benchmarks in headless Chrome 148 on an Apple M4 Max laptop, all times in seconds. Measurements come from the `chrome-routing-bench` example, which runs the unmodified browser bundle of GJS. *Routing* sums the CDT-construction and Dijkstra-tree sleeve-search phases of Section 4.1 on the full graph; *Tiling* is the build of the tile pyramid, capacity $\mathcal{C} = 500$ with Z chosen by the natural stops of Section 3, which includes the per-level routing; *Total* is the total loading time including parsing of the graph file, graph layout, and all other stages.

Graph	$ V $	$ E $	Routing	Tiling	Total
gameofthrones	407	2 639	0.07	0.07	0.38
composers	3 405	13 832	1.49	0.74	2.69
ca-GrQc	5 242	28 968	1.78	0.87	3.40
facebook_combined	4 039	88 234	4.34	2.36	7.77
ca-HepTh	9 877	51 946	9.33	3.41	14.87
ca-HepPh	12 008	236 978	38.89	14.08	57.71
ca-CondMat	23 133	186 878	79.72	20.52	110.06
deezer_europe	28 281	92 752	134.78	23.09	172.68
delaunay_n15	32 768	98 274	18.71	5.64	41.38

for those nodes, subject to the geometric limitations described in Sections 3.2 and 4.2.

Interaction. The renderer supports pan, zoom, and hover highlighting of nodes and edges. Hovering an edge highlights it and its incident nodes; hovering a node highlights the node, its one-hop neighbors, and its two-hop neighbors, each in a different color. A `zoomTo` API animates the viewport to a target rectangle with a smooth transition and is used, for example, by the search box to focus the view on a query result.

6 EXPERIMENTAL RESULTS

We evaluate the sleeve router and the tile-pyramid build on the same benchmark graphs as Table 1: the Game of Thrones social network [13], the GD 2011 contest composers graph [9], four SNAP collaboration networks (ca-GrQc, ca-HepTh, ca-HepPh, ca-CondMat) and the SNAP facebook_combined ego network [39, 3], the deezer_europe social network [44], and the SuiteSparse Delaunay mesh delaunay_n15. The loading and routing-mode experiments run in headless Chrome 148 on an Apple M4 Max laptop with 48 GB of unified memory, with the unmodified browser bundle of GJS loaded into the page. Layout uses IPSep-CoLa, and the loading benchmark builds the tile pyramid with the same parameters as Table 1: capacity $\mathcal{C} = 500$, with Z chosen by the natural stops of Section 3.

Table 2 reports per-graph timings of three pipeline stages: *Routing*, the sum of the two phases logged by the sleeve router on the full graph—constrained Delaunay triangulation of the obstacles and the Dijkstra-tree search on the CDT dual, Section 4.1; *Tiling*, the build of the tile pyramid, Section 3, which includes the per-level routing; and *Total*, the end-to-end loading time including parsing.

CDT construction takes about 6 ms on Game of Thrones and at most 420 ms in this benchmark. It accounts for about 8% of *Routing* on Game of Thrones and at most 6.4% on the other graphs. The remaining time is spent in the routing pass, which includes sleeve search and path extraction, with the vertex-cover heuristic of Section 4.1 reducing the number of trees. The full pipeline finishes in under three minutes on all nine graphs; this is a one-time cost for a fixed layout. The resulting tile pyramid already contains the per-level routes, so panning and zooming fetch and render precomputed tiles without rerouting.

Table 3 isolates the contribution of the two source-side optimizations of Section 4.1: batched Dijkstra trees over per-edge A^* and the vertex-cover heuristic on top of that. Each row times a single sleeve-search pass run after layout, with the constrained De-

Table 3: Sleeve-search times of the three routing modes on the benchmark graphs of Table 2, in headless Chrome 148. Columns: *Srcs*, the number of distinct source endpoints in the original edge orientation, used to group DJ queries (A^* still searches per edge); *VC*, the number of source groups after the greedy maximum-degree vertex-cover reorientation of Section 4.1; A^* , wall-clock seconds of one full sleeve-search pass with per-edge A^* (straight-line distance to the target as heuristic); *DJ*, wall-clock seconds with the batched Dijkstra-tree mode—one tree per source, expanding until all of that source’s targets are reached; *VC.DJ*, wall-clock seconds with the vertex-cover-reoriented Dijkstra-tree mode—same as DJ but with only VC roots (groups resolved entirely by straight-line tests need no tree); *DJ red.* = $(A^* - DJ)/A^*$, relative time reduction of DJ over A^* ; *VC red.* = $(DJ - VC.DJ)/DJ$, additional time reduction of VC.DJ over DJ. The CDT-construction phase, identical across modes, is excluded so that only the search costs are compared.

Graph	Srcs	VC	Time (s)		Time reduction		
			A^*	DJ	VC.DJ	DJ red.	VC red.
gameofthrones	331	198	0.077	0.059	0.043	+23 %	+28 %
composers	2 645	1 281	2.60	2.14	1.49	+18 %	+31 %
ca-GrQc	5 241	2 796	2.55	2.52	1.65	+1 %	+35 %
facebook_combined	3 663	3 045	13.31	4.88	4.36	+63 %	+11 %
ca-HepTh	9 875	5 003	17.13	15.42	9.28	+10 %	+40 %
ca-HepPh	12 006	7 032	111.2	58.07	36.38	+48 %	+37 %
ca-CondMat	23 133	13 564	189.1	131.1	76.45	+31 %	+42 %
deezer_europe	21 060	13 450	172.1	154.0	124.4	+11 %	+19 %
delaunay_n15	28 467	23 456	13.57	16.02	14.88	-18 %	+7 %

launay triangulation excluded so that only the search costs are compared. The batched Dijkstra trees outperform per-edge A^* on every social graph tested, reducing elapsed time by up to 63 % on *facebook_combined* and 48 % on *ca-HepPh*; the vertex-cover heuristic yields a further 11–42 % reduction across the social graphs. On the Delaunay mesh *delaunay_n15*, batching alone is 18 % slower than A^* . Vertex-cover reorientation recovers 7 % relative to batching alone, but remains about 10 % slower than A^* . Low degree and limited shared search regions are plausible explanations, not separately measured causes.

Browsing smoothness. We test how smoothly a user can browse a freshly loaded graph in the browser. On the same M4 Max laptop as Tables 2 and 3, a script loads a graph, waits for the tile pyramid to build, and then runs two trials of three dives each (six dives per graph). Each dive picks a random point inside the laid-out graph, recenters the view on that point, zooms from the coarsest view all the way in to the finest detail in one second, then zooms back out to the coarsest view in another second. The three dives in each trial run back to back. These browsing runs use separately built pyramids; their recorded level counts are reported in Table 4 and differ from the natural-stop builds in Table 1. The two tables should therefore not be read as measurements of identical tile pyramids.

While the dives play, the script measures three things on the browser side. First, the interval between successive animation-frame callbacks, whose nominal spacing at 60 Hz is 16.7 ms. We report the arithmetic mean of the two trial-specific 95th percentiles, not a pooled percentile or a direct measurement of displayed frames. Second, the number and total duration of main-thread tasks lasting more than 50 ms, summed over both trials. Such “long tasks” can delay rendering and input handling. Third, at the deepest point of every dive, how many graphical objects the browser must draw, that is, nodes, edge clips, labels, and arrowheads, summed over visible tiles intersecting the viewport. We report the maximum of these six samples. This count is a proxy for rendering load, not a GPU-time measurement or a maximum over all frames. Table 4 reports the result.

Table 4: Browsing-smoothness measurements on the benchmark graphs. For each graph the script runs two trials of three dives; each dive picks a random point inside the laid-out graph, recenters on it, and zooms from the coarsest view to the finest and back in two seconds. *Levels*, number of tile-pyramid levels built ($Z + 1$); *LT*, number of main-thread tasks longer than 50 ms summed over both trials; *LT total*, their total duration, in milliseconds; *Frame p95*, the mean of the two trial-specific 95th percentiles of animation-frame callback intervals, in milliseconds; *Elts max*, the largest number of graphical objects, that is, nodes, edge clips, labels, and arrowheads, that the browser had to draw at the deepest point of any dive.

Graph	V	E	Levels	LT	LT total	Frame p95	Elts max
gameofthrones	407	2 639	5	0	0	17.4	2 567
composers	3 405	13 832	9	1	51	18.2	497
ca-GrQc	5 242	28 968	9	0	0	18.2	677
ca-HepTh	9 877	51 946	9	0	0	18.2	383
facebook_combined	4 039	88 234	9	0	0	18.0	495
ca-HepPh	12 008	236 978	9	9	556	18.4	1 394
ca-CondMat	23 133	186 878	9	5	273	18.2	1 391
deezer_europe	28 281	92 752	9	3	164	18.4	2 505
delaunay_n15	32 768	98 274	9	0	0	18.1	1 012

The mean trial-specific frame-interval percentiles are 17.4–18.4 ms after rounding, close to the nominal 16.7 ms interval at 60 Hz. These statistics do not rule out longer stalls or establish perceptual smoothness. There are nine, five, and three long tasks on *ca-HepPh*, *ca-CondMat*, and *deezer_europe*, respectively, one on *composers*, and none on the remaining graphs. At the sampled deepest zooms, the browser is asked to draw at most 2 567 objects on *gameofthrones* and at most 2 505 on *deezer_europe* even though the latter graph has 35 times more edges. This is the property the pyramid is designed to deliver: the draw workload is tied to visible tile contents rather than requiring the full graph to be drawn each frame. This is an empirical observation, not a graph-size-independent complexity bound.

Evaluation scope. The loading and routing-mode tables report individual runs, not repeated-run means with confidence intervals. The browsing experiment covers only six sampled locations per graph on one machine, and some samples contain no graph entities. It does not establish worst-case frame times, performance on lower-end hardware, or user task effectiveness. We also lack a controlled comparison with non-tiled rendering and a quantitative measure of visual stability across level changes.

7 CONCLUSION

We presented GJS, an open-source TypeScript library for graph layout, edge routing, and visualization in web browsers. The sleeve routing method routes edges directly on the CDT dual graph using batched Dijkstra trees and the funnel algorithm. The tiling scheme enables map-like exploration of large graphs with level-of-detail filtering and edge simplification. The WebGL renderer, built on *deck.gl*, provides smooth pan-and-zoom interaction.

Future work. The current approach builds the entire tile pyramid up front, which is expensive and bounds the size of graphs we can handle. A natural next step is to build only the first few levels quickly and hand them to the renderer; the remaining levels would be built lazily as the user browses the graph. This could lower the time-to-first-frame and reduce resident tile storage, but layout, routing, and graph storage would still limit graph size.

A second challenge is that some tiles can contain many more elements than the capacity parameter $\mathcal{C}$ that guides the build. Table 1 reports densest tiles holding approximately $5.9\text{--}7.3\times$ the nominal $\mathcal{C}$ in the worst cases, which leaves the rendering and interaction

cost uneven across the viewport. Enforcing an actual per-tile budget remains an open problem.

SUPPLEMENTAL MATERIALS

GJS is available as open source; an anonymized mirror of the repository is provided for review at <https://anonymous.4open.science/r/abcde-386C/>. A live demo of the WebGL renderer with sleeve routing is available at <https://gjs-sleeve.surge.sh/>. The benchmark drivers used for Tables 1–4 ship with the repository.

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